Session Objectives

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1. Understand how **anatomic/structural changes & wounds impact the physiologic function or dysfunction** that may produce/promulgate **somatic dysfunction** (SD). Compare and contrast internal vs. external considerations in health and disease in Gynecology - highlight structural considerations at different life stages and how osteopaths can be of service.
2. Correlate importance of perfect **Circulation** in health and disease of gynecologic concerns - Understand the **role of thorax, diaphragm & lymphatics** for the female patient whether for reproductive organs or otherwise. Understand the osteopathic physicians' role and responsibility with regard to "the want of supply and the burden of dead deposits"
3. Describe the importance of healthy balance of **Nervous system** in health & disease of gynecologic concerns. Understand importance of the **autonomic nervous system** (and "**facilitation"** of related segments) with respect to reproductive function and the whole patient. Discuss what **spinal levels or craniosacral levels** are involved with **autonomics & circulation** to/ from the GU system**
4. Discuss Osteopathic Considerations of Common Gynecologic Complaints.
5. Review and Describe the genitourinary **Chapman's points**

Where to start looking for Health?

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- Diaphragm (nerve and arterial supply)
- Pancreas, liver, GB, spleen:
  - Derangement related to
    - Dyspepsia, dropsy, enlarged spleen, engorged liver, cancer, gall-stones, skin eruptions, change of color, constipation, inflammatory diseases, ulceration of the stomach, bowels, kidneys and uterus.
- Remove embarrassments from blood and nerve supply which could be caused by: strains, jars, nervous shocks, or wounds produced by change of season, climate, physical injuries of all kinds, be they great or small.

Begin at:

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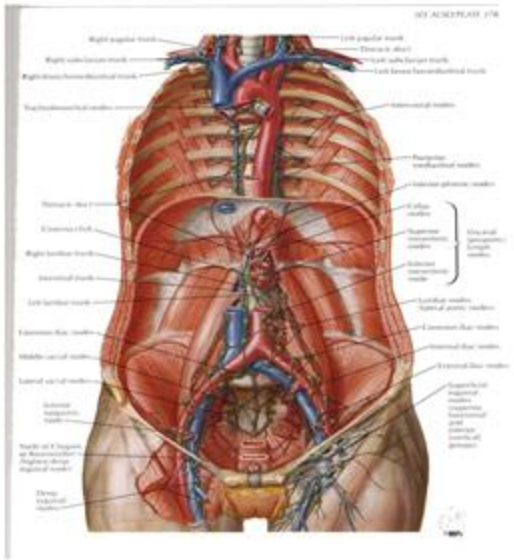
- The bones and know them, their uses and their places
- Then the binding ligaments…
- In particular her spine and pelvis…
- Search until we know what effect would be produced on any organ of the chest or abdomen by an anterior, posterior or lateral change of the atlas, axis or other bones of the neck;
- What effect on the heart would follow the changed position of a rib…?
- T1-T12, lumbar, sacrum innominates, bc of a slip or change from the normal. Would it send more or less blood to the brain?...

Considerations Continued

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- Lungs to suffer loss of power purifying blood and then a failure in the power of the heart would appear,…
- Then a stagnation … liver, spleen pancreas….
- Ad infinitum…congestion and sluggish action in the organs upon which a healthy womb has to depend for its vitality and functioning….”



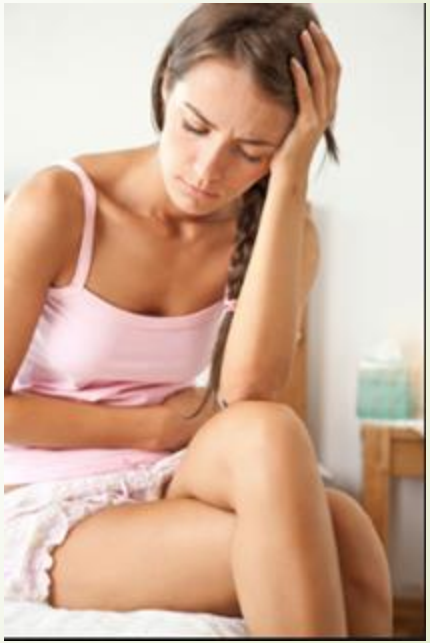
More thoughts from Dr Still P&MPO text

Gynecologic Considerations

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- Spine (Exaggerated curves: A-P, lateral)
- Pelvic Floor Dysfunction
- Dysmenorrhea (primary or secondary)
- Pre-menstrual syndrome
- Breast Heaviness/ Pain
- Pelvic Congestion
- Myomata (Fibroids)
- Leucorrhea
- Pelvic Inflammatory Disease
- Endometriosis
- Adhesions
- Ovarian cysts/ remnants
- Tumefaction
- Dyspareunia
- Infertility
- Breast



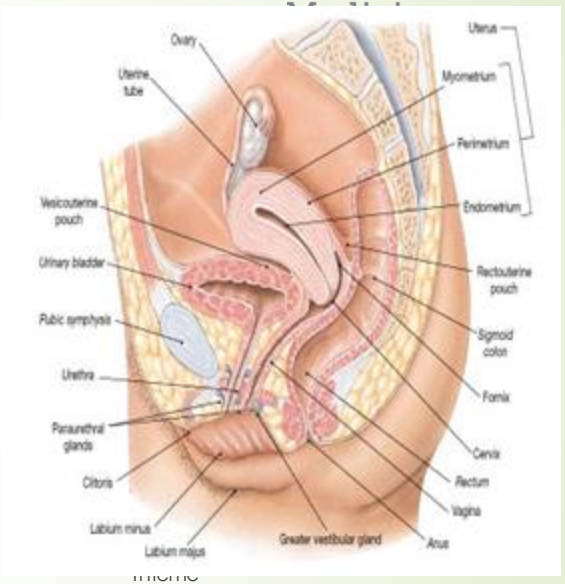
Structure and Function are Interrelated

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Consider Impact of:

- Wounds
- Crowding
- Postural issues
- Nutrition
- Exercise or lack thereof
- Surgical scars/ adhesions
- Scoliosis
- Sympathotonia
- Inflammation
- T.A.R.T.
- Gestation
- External stressors- tight clothing
- Psychological or physical trauma
- Impeded/Hindered blood supply
- Impeded/Hindered drainage



*Importance of Thoraco-Lumbar (T-L) Junction

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- An area of MSK Strain

- **Good motion T-L important for proper Diaphragmatic excursion!**
  - Crus of posterior diaphragm attachment L1-3 or L4
  - Think systemically: lymphatic and venous drainage

- **T10-L2** = SNS for **bowel, uterus and ovaries/ adnexa, bladder and ureters**, adrenals and renal
  - Any facilitation at these somatic segments by one organ can affect other organs supplied by the same segments

- *Exacerbation of facilitation of this area is common with MSK TART changes*

Impact of Wounds or TART on Ovaries

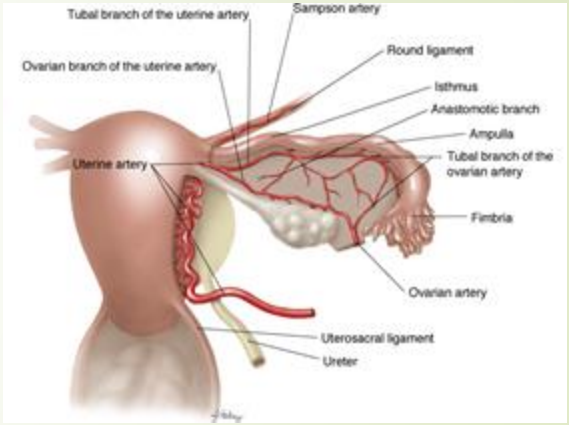
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Arterial Supply: Ovarian artery
from aorta ~level of **L2**, Anastomose with branches of Uterine artery.

Venous Drainage:
Right: ovarian vein to the IVC
Left: to renal vein

Lymph Drainage: peri-aortic

SNS: T10-11
PSNS: Vagus



"*Mittelschmerz" referred through afferents along T10 dermatome – periumbilical area*

NOTE: Ovary often close to appendix and shares innervation w/ kidney

Impact of Wounds or TART upon Uterus

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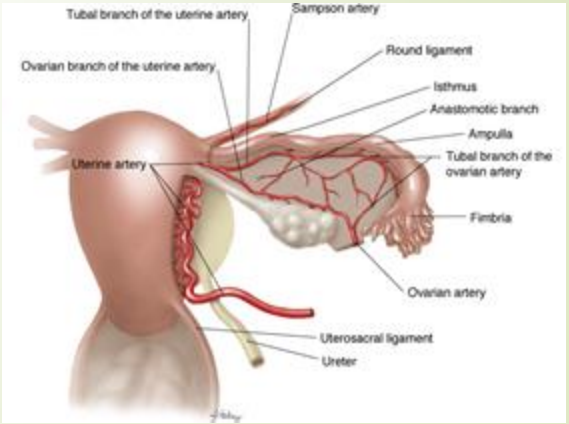
Arterial Supply: Uterine (from Internal Iliac) and Ovarian arteries

Venous Drainage: Uterine venous plexus
Connected with Superior rectal v.
Forms Portal systemic anastomosis

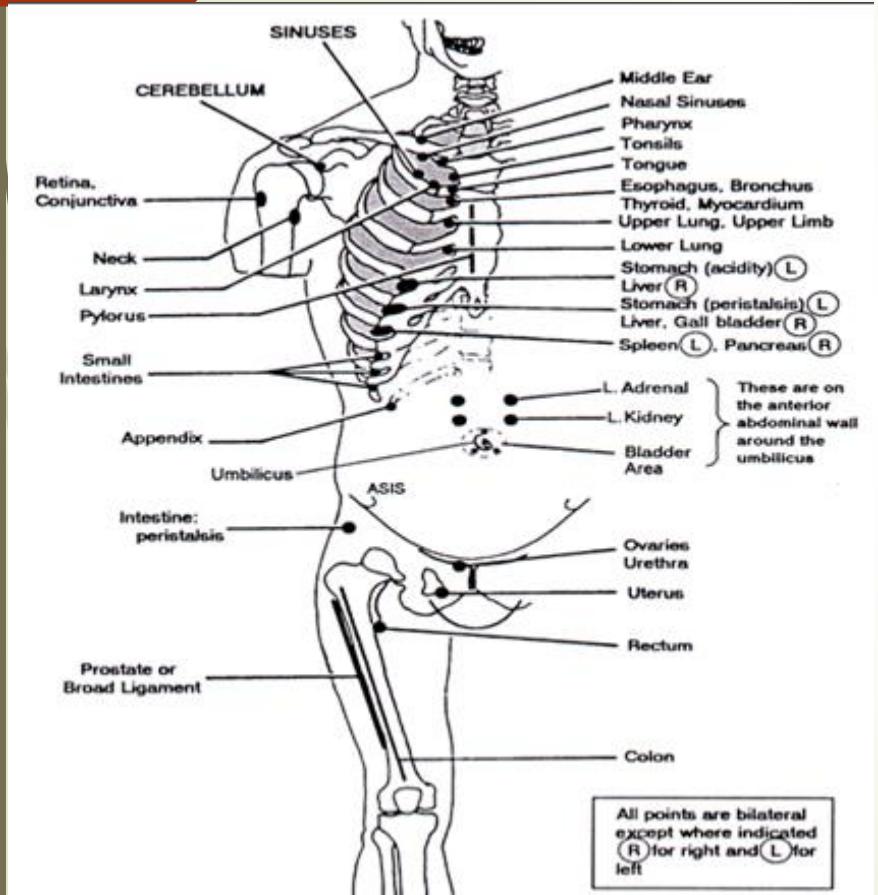
Lymph Drainage: 3 main routes: Aortic, Internal & External Iliac Nodes

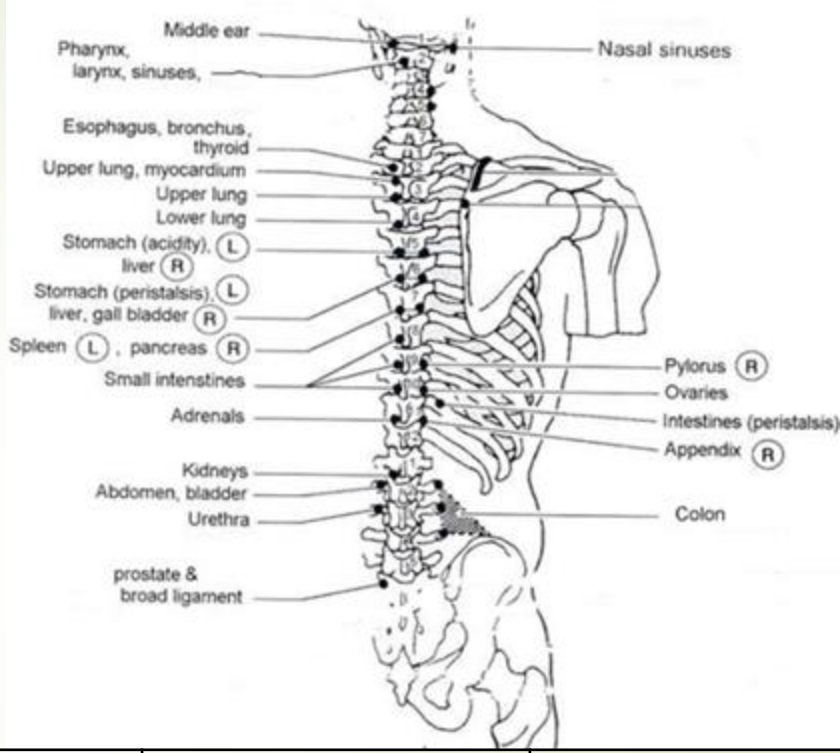
SNS: Inferior hypogastric plexus to T11-L2 spinal levels (Inferior Mesenteric Ganglion)

PSNS: S2 – S4



Chapman's Points*





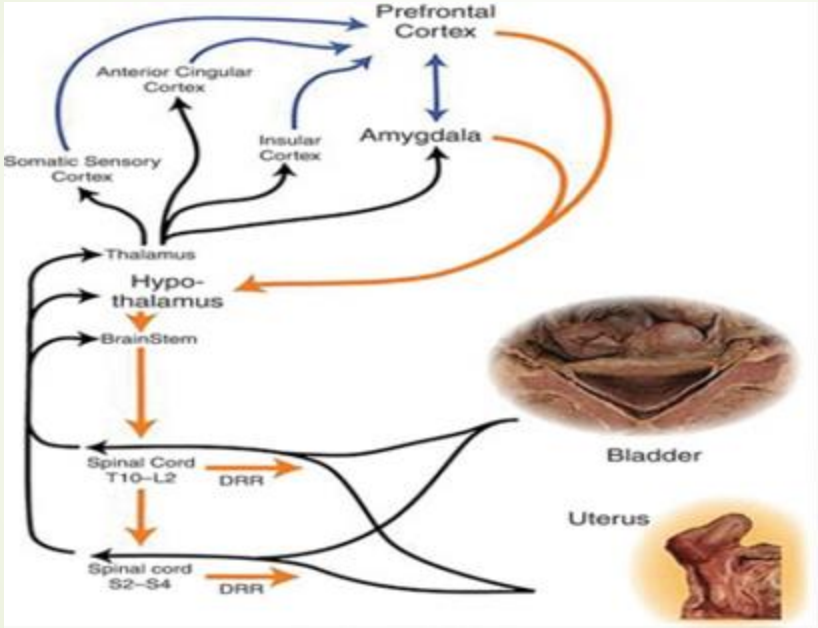
Organ	Anterior	Posterior
Ovaries	Pubic Tubercle	T10 TP
Uterus	Inferior pubic rami	Lateral sacral base
Broad Ligament	Posterior-Lateral IT band	PSIS

LWW Health Library

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From: **Gynecology**

Foundations of Osteopathic Medicine: Philosophy, Science, Clinical Applications, and Research, 4e, 2018



Legend:

Diagram showing innervation of the uterus and bladder. Note that these structures can receive dorsal root reflexes (DRRs) through the afferent fibers. This gives rise to **cross-sensitization where dysfunction of one organ can influence the surrounding organs.** This principle makes localization of visceral pain challenging.

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Breast: Lymphatic Drainage

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Lymphatic movement

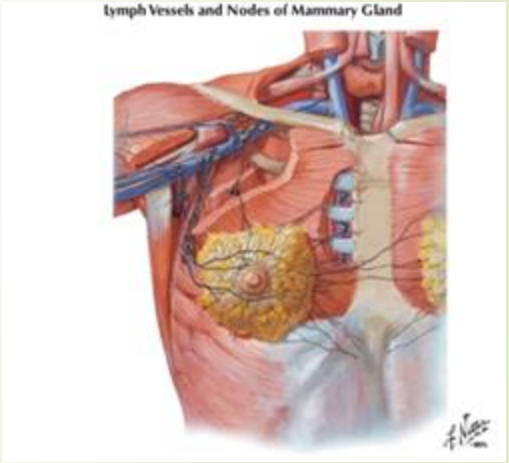
- Diaphragms
- Abdominal lifts

Scapular release

- Help swollen breasts and tenderness
- Encourage milk let down

Mastodynia

- *T1-8 (especially T4-6) Somatic dysfunction*
- *Anterior/posterior ribs*
- *Thoracic Outlet*
- *Zink treatment of breast*
- *Upper Extremity distraction*



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"Your duty as a master mechanic is to know the engine is kept in perfect condition, so that there will be no functional disturbance to any nerve, or vein, or artery that supplies ad governs the skin, the fascia, the muscle the blood or any fluid that should be in free circulation to sustain life and renovate the system from deposits that would cause what we call disease."

- AT Still - Philosophy and Principles of Osteopathy pg 70





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Osteopathic Treatment: What makes You special

Your ability to RECOGNIZE, DIAGNOSE & TREAT “Somatic dysfunctions” and to INTEGRATE this ability into the care of your patients to help optimize their health.

OMT
Treat what you find.
Be sure to check:

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Transition zones/ junctions
O-A, C-T, T10-L2, L-S, SI's

Sacral mobilization/ rocking

Lymphatic drainage
Axillary folds, diaphragms,
Thoracic Inlet and cage motion, mesenteric lifts…

Cranial
SBS, C2, OM Suture …

Question

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A 22-year-old female presents to the office complaining of pain and cramping with her menses. She states the pelvic and back pain typically starts 1-2 days before the onset of her menses. + nausea and fatigue. On physical examination, you note a well-appearing female with a normal abdominal and pelvic examination. Structural examination will most likely reveal a sympathetic viscero-somatic reflex at:

A – OA, C2
B – T1-4
C – T11-L2
D – L3- L5
E – S2-4

Question

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A 17-year-old overweight female presents to your office complaining of irregular and painful menses. You note on physical exam, increased acne and facial hair. Pelvic examination reveals an increased density/ areas of swelling around the ovaries. A) What do you suspect and B) Where would you most likely find a Chapman's point related to this patient's presentation?

A- Right T12 Transverse Process

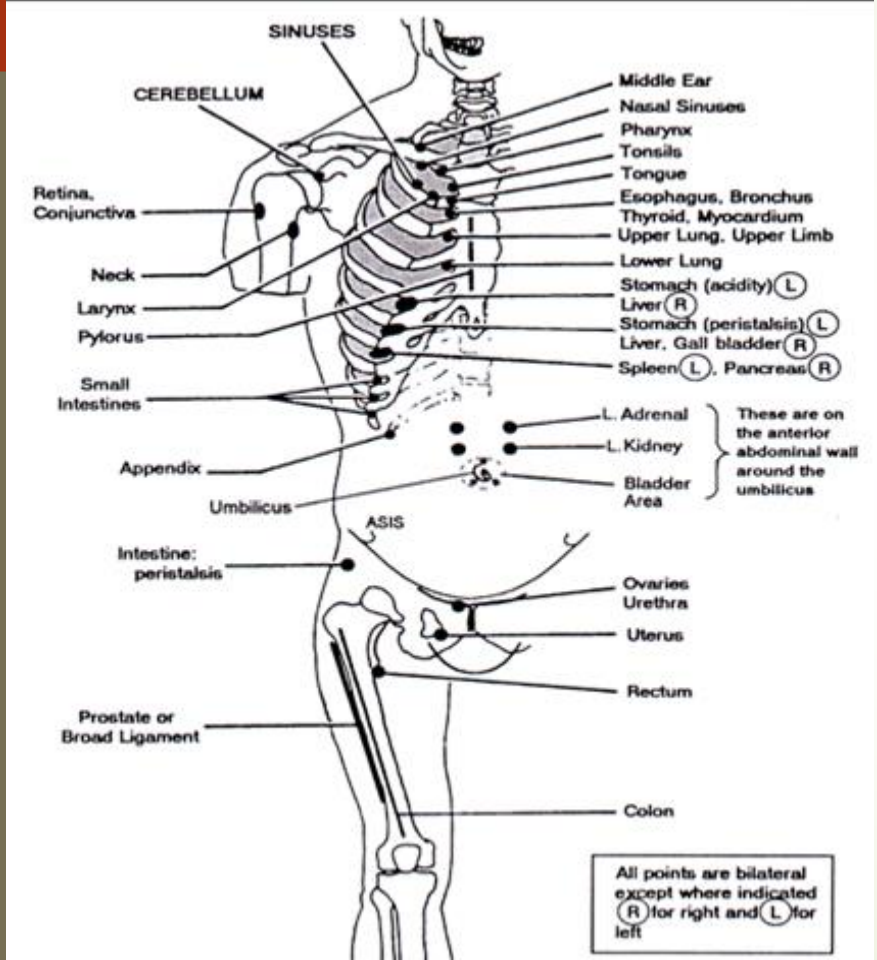
B –Lateral to the pubic symphysis on the Pubic tubercle

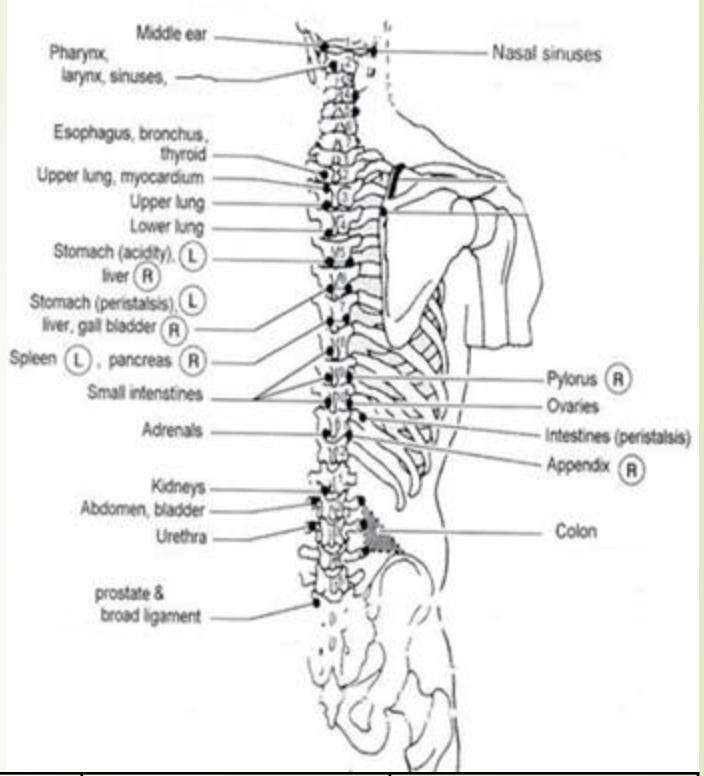
C – Upper Medial Thigh

D – Inferior pubic ramus

E – Posterior L3 Transverse Process

Chapman's Points





Organ	Anterior	Posterior
Ovaries	Pubic Tubercle	T10 TP
Uterus	Inferior pubic rami	Lateral sacral base
Broad Ligament	Posterior-Lateral IT band	PSIS

Summary Slide

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"It matters not whether the cause is far remote or in close proximity to the uterus; we must find it, or we will be found in the antediluvian tribe of speculum cranks of all the blind female doctors' ages."

-Andrew Taylor Still
Discoverer of the Science of Osteopathy

Pg 207 on the Pelvis P&MPO